Aly Morsy

Neufahrn b. Freising, Germany | aly.morsy@tum.de | aly.morsy@volkswagen.de | +49 15257560590 | linkedin.com/in/ali-nasser-229304143

Summary

Computational engineer specializing in computational materials science, atomistic simulation, scientific machine learning, and multiscale modelling. Currently conducting Master's thesis research at Volkswagen AG on first-principles-informed atomistic spin dynamics of rare-earth permanent magnets using Density Functional Theory (DFT), Monte Carlo methods, and Landau-Lifshitz-Gilbert spin dynamics. Additional research experience includes physics-informed deep learning, AI surrogate modelling, uncertainty quantification, and high-performance scientific computing for complex physical systems. Interested in developing AI-driven computational methods for accelerated materials discovery.

Core Skills

Computational Materials Science: Density Functional Theory (DFT), Atomistic Spin Dynamics, VAMPIRE, Monte Carlo Methods, Constrained Monte Carlo, Landau-Lifshitz-Gilbert Dynamics, Magnetic Materials, Multiscale Modelling

Artificial Intelligence: Scientific Machine Learning, Deep Learning, TensorFlow, Neural Networks, Physics-Informed Learning, Surrogate Modelling, Bayesian Inference, Gaussian Processes, Principal Component Analysis (PCA), Probabilistic PCA

Programming: Python, MATLAB, Fortran, Linux/Unix, Shell Scripting, HPC Workflows, Numerical Modelling, Scientific Computing

Engineering Software: VAMPIRE, OpenFOAM, MFiX, ANSYS CFX, ParaView, CATIA V5, SolidWorks

Experience

Volkswagen AG — **Wolfsburg, Germany**

Master Thesis Student – Computational Materials Modelling — *Apr. 2026 – Present*

- Develop multiscale computational workflows linking Density Functional Theory (DFT), atomistic spin dynamics, and continuum-scale magnetic properties of rare-earth permanent magnets for electric vehicle applications.
- Perform atomistic spin dynamics simulations of $Nd_2Fe_{14}B$ using Monte Carlo, constrained Monte Carlo, and Landau-Lifshitz-Gilbert spin dynamics implemented in VAMPIRE.
- Predict finite-temperature magnetic properties including magnetisation, magnetic anisotropy, exchange stiffness, and domain-wall behaviour.
- Investigate machine-learning approaches for predicting quantum-to-classical temperature rescaling relationships in atomistic spin dynamics, aiming to improve the transferability of computational magnetic material models.
- Conduct large-scale simulations and automated post-processing on Linux HPC systems using Python and shell scripting.

Bayer AG / Politecnico di Torino — **Leverkusen, Germany / Turin, Italy**

Industry Practicum – Scientific Machine Learning — *Oct. 2024 – Mar. 2026*

- Developed a physics-informed 3D U-Net deep learning model for predicting scalar transport fields from high-fidelity simulation datasets.
- Generated and processed OpenFOAM datasets using homogeneous isotropic turbulence to train surrogate models for computational physics.
- Applied scientific machine learning and physics-informed loss functions to accelerate computationally expensive numerical simulations.

Ecole des Mines de Saint-Etienne — **Saint-Etienne, France**

Research Intern – Computational Modelling — *Aug. 2025 – Sept. 2025*

- Worked on numerical modelling of coupled particle-fluid systems using MFiX and Fortran.
- Developed experience in multiphysics modelling, numerical methods, and scientific computing.

Helwan University **Cairo, Egypt**

Teaching Assistant and Researcher – Mechanical Engineering *Sept. 2023 – Sept. 2024*

- Supported teaching activities in Fluid Mechanics while assisting undergraduate students in numerical and engineering analysis.

Elaraby Group **Benha, Egypt**

Research and Development Mechanical Engineer *Dec. 2021 – Aug. 2023*

- Contributed to product development projects involving thermal management, refrigeration, thermoelectric cooling, and sustainable engineering solutions.
- Worked within multidisciplinary engineering teams on product design, prototyping, technology scouting, and patent-related activities.

Research Highlights

AI-Assisted Temperature Rescaling in Atomistic Spin Dynamics **Volkswagen AG**

- Investigate machine-learning approaches to predict the mapping between classical atomistic spin dynamics temperatures and quantum-informed physical temperatures.
- Aim to reduce reliance on empirical temperature-rescaling calibration and improve the transferability of spin models to new magnetic materials.

Physics-Informed Deep Learning for Scientific Simulation **Bayer AG / Politecnico di Torino**

- Built a 3D U-Net surrogate model to predict scalar transport fields from simulation-generated datasets.
- Used TensorFlow, Python, OpenFOAM data, and physics-informed loss functions to improve prediction consistency and accelerate numerical simulations.

Education

Erasmus Mundus Joint Master in Multiphase Systems for Sustainable Engineering **2024 – 2026**

Ecole des Mines de Saint-Etienne, Politecnico di Torino, Technical University of Munich *France, Italy, Germany*

- Focus: computational modelling, multiscale systems, uncertainty quantification, scientific computing, hydrogen technologies, and sustainable engineering.
- Relevant coursework: Multiscale Modelling, Mathematical Modelling of Materials, Probability Theory and Uncertainty Quantification, Bayesian Inference, Gaussian Processes, Principal Component Analysis, Probabilistic PCA, Computational Thermo-Fluid Dynamics, Multiphase Flow in Engineering Systems, Hydrogen for Sustainable Energy.

Bachelor of Science in Mechanical Power Engineering **2016 – 2021**

Helwan University, Faculty of Engineering *Cairo, Egypt*

- Final grade: Very Good with Honors. Graduation project: performance study of brackish water desalination using reverse osmosis powered by solar photovoltaics.

Languages

Arabic: Native | English: Full Professional Proficiency | German: Professional Working Proficiency

References

Dr. Jonathan Edward Mueller
Advanced Energy Materials
Battery and Fuel Cell, Volkswagen AG – K-GERS/F
Master Thesis Supervisor
jonathan.edward.mueller@volkswagen.de

Prof. Daniele Marchisio
Professor
Department of Applied Science and Technology (DISAT)
Industry Practicum Supervisor
daniele.marchisio@polito.it